Correction of the first interrogation of Algebra I
November 2023 (75 mn)

Exercice 1 .(06 Pts)
Let f be an application of $\mathbb{R}$ in $\mathbb{R}$. Deny, as precisely as possible, the following statements :

1. For all $x \in \mathbb{R}$ $f(x) \leq 1$.**(01 pts)**
2. The application f is increasing.**(01 pts)**
3. The application f is increasing and positive.**(01,5 pts)**
4. There exist $x \in \mathbb{R}^+$ such that $f(x) \leq 0$.**(01 pts)**
5. There exist $x \in \mathbb{R}$ such that for all $y \in \mathbb{R}$, if $x < y$ then $f(x) > f(y)$.**(01,5 pts)**

Exercice 2 .(06 Pts)
Soient $A, B, C \in P(E)$. Et soient f et g deux applications de A dans B et h une application injective de B dans C.

1. Montrer que $A \cup B = A \cap C$ si et seulement si $B \subset A \subset C$. **(04 pts)**

2. Montrer que $h \circ f = h \circ g \Rightarrow f = g$. **(02 pts)**

Exercice 3 .(08 Pts)
Let the applications f and g defined by :

$$f : \begin{array}{ccc} \mathbb{R}^2 & \longrightarrow & \mathbb{R} \\ (x,y) & \longrightarrow & xy \end{array} \quad \text{et} \quad g : \begin{array}{ccc} \mathbb{R} & \longrightarrow & \mathbb{R}^2 \\ x & \longrightarrow & (x,x^2) \end{array}$$

1. Calculate $f \circ g$ and $g \circ f$ (precise the domain and the codomain). **(02 pts)**

2. let the sets A = $\left\{(1,2), \left(\sqrt{2}, \frac{2}{\sqrt{2}}\right)\right\}$ and B = $\{(2,-7)\}$. Calculate $f(A) and\ g^{-1}(B)$.

3. Is f injective ? is g surjective ? **(02 pts)**

Bon courage